# SYSTEM DESIGN

## 1 Introduction

This chapter describes the overall system architecture and data flow of the **ResQ-DMS (Disaster Management System)**. The system is designed as a cloud-based web application that supports disaster preparedness, monitoring, response, and coordination during emergency situations.

The system integrates multiple modules such as disaster monitoring, risk analysis, volunteer coordination, shelter management, and missing person tracking. These modules work together to provide real-time information and support disaster response operations.

Data Flow Diagrams (DFDs) are used to illustrate how data moves through the system, how it is processed, and how it interacts with external entities and internal databases.

The system architecture focuses on modular structure, security, reliability, and efficient data management.

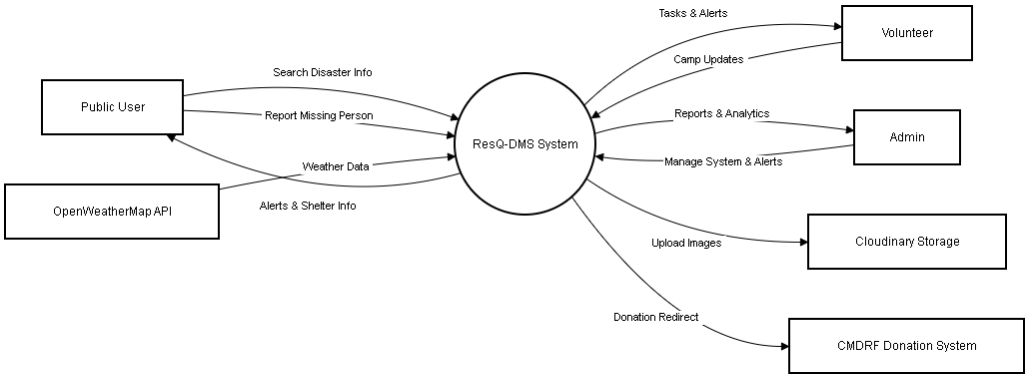
## 2 Context Flow Diagram

A Context Flow Diagram represents the entire system as a single process and illustrates the interaction between the system and external entities.

In the ResQ-DMS system, external entities include public users, volunteers, administrators, and external services such as weather APIs and cloud storage. The context diagram shows how these entities exchange information with the system.

Public users interact with the system to access disaster alerts, search for shelters, report missing persons, and make donations. Volunteers assist in updating disaster-related information and managing relief camps. Administrators monitor the system, approve volunteers, and broadcast emergency alerts.

External services such as the OpenWeatherMap API provide weather information, while Cloudinary is used for storing images related to missing persons and shelters. Donations are processed through the Chief Minister's Distress Relief Fund (CMDRF) system.



## 3 Data Flow Diagram

A Data Flow Diagram (DFD) is a graphical representation used in system analysis and design to illustrate how data moves through a system.

DFDs help in understanding how information flows between different components of the system. They show how data enters the system, how it is processed, where it is stored, and how it is delivered to users or external systems.

The main components of a Data Flow Diagram include:

**External Entities**
External entities represent sources or destinations of information outside the system. Examples include users, administrators, and external services.

**Processes**
Processes transform input data into output data. They represent operations performed within the system.

**Data Stores**
Data stores represent repositories where system data is stored. These may include databases or files.

**Data Flow**
Data flow represents the movement of data between entities, processes, and data stores.

DFDs are usually developed in multiple levels to provide both high-level and detailed views of the system.

## 4 Rules for DFD Construction

The following rules are followed while constructing Data Flow Diagrams:

- Begin with a high-level context diagram.
- Clearly identify all external entities.
- Label each data flow and process clearly.
- Avoid including implementation details such as algorithms or program logic.
- Ensure consistency between different levels of DFDs.
- Decompose complex processes gradually into lower levels.

## 5 DFD Symbols

The following symbols are used in Data Flow Diagrams:

**Process -** ○
A process represents a transformation of input data into output data.

**Data Store - ‖**
A data store represents a storage location where system data is kept.

**Data Flow - →**
Data flow represents the movement of data between processes, entities, and data stores.

**External Entity - □**
External entities are outside systems that interact with the application.

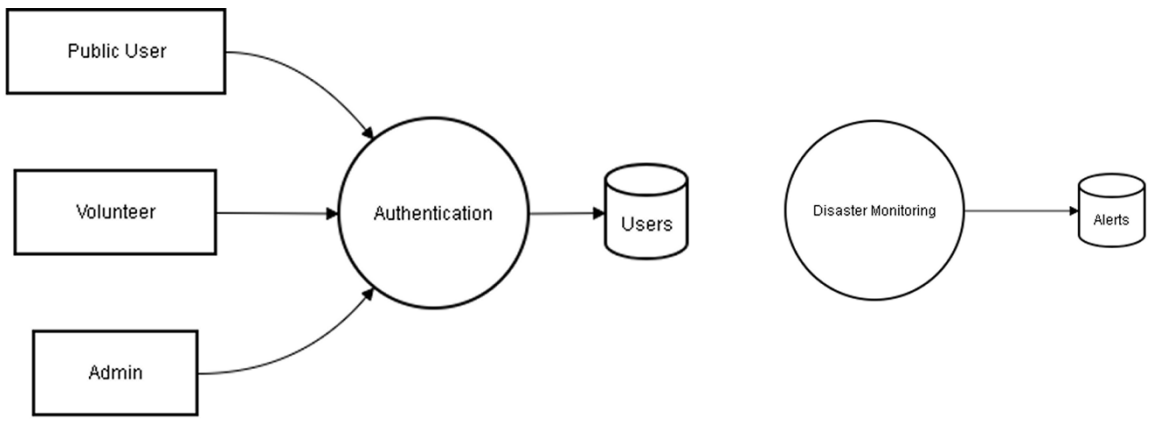
## 6 DFD Level 1 – ResQ-DMS System

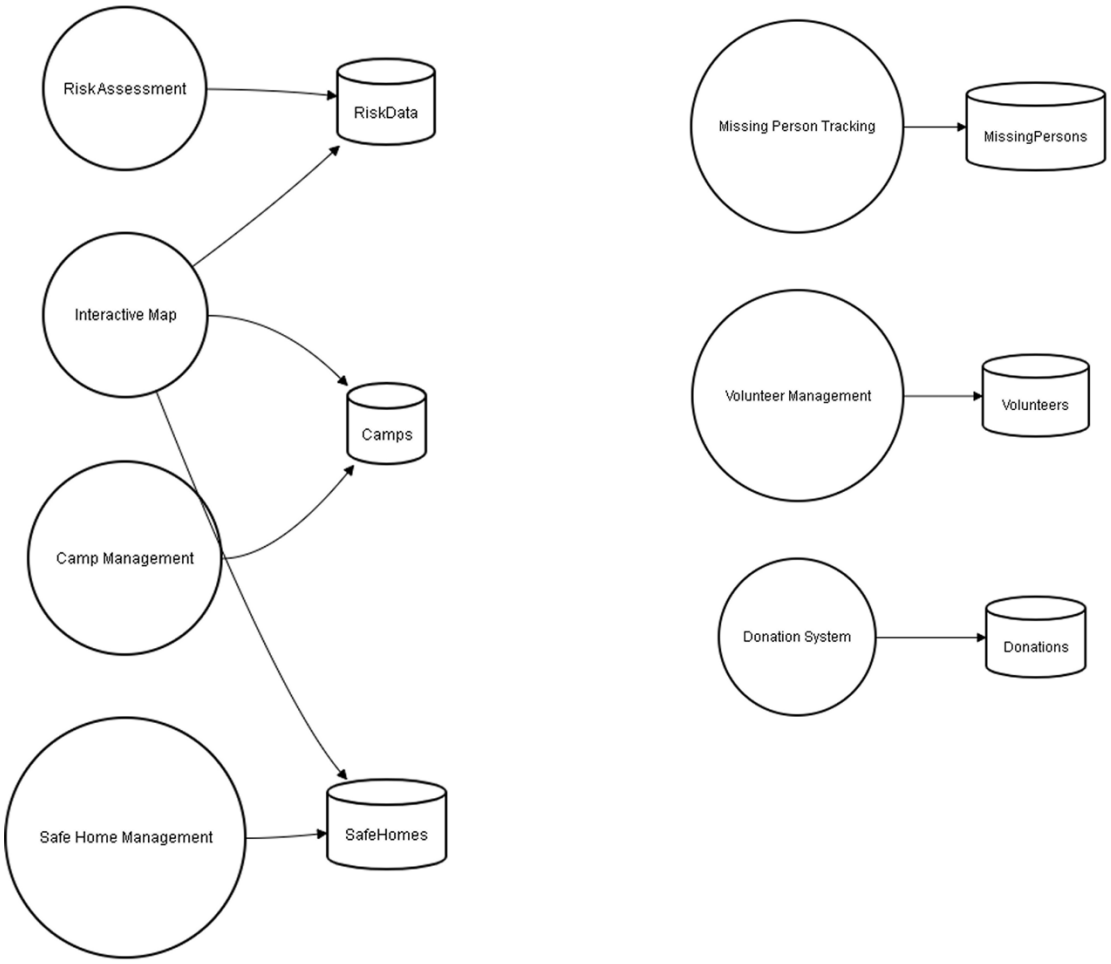
The Level 1 Data Flow Diagram decomposes the entire system into major functional modules.

The ResQ-DMS system consists of the following modules:

1. Authentication and Role Management
2. Disaster Monitoring and Alert System
3. Risk Assessment and Analytics
4. Interactive Disaster Map
5. Camp Management System
6. Safe Home and Shelter Management
7. Missing Person Tracking System
8. Volunteer Management System
9. Donation Management System

Each module interacts with the system database and external services to process and manage disaster-related information.






## 7 DFD Level 2 Diagrams

Level 2 Data Flow Diagrams provide a detailed view of individual modules within the system. Each module is broken down into smaller processes that show how data is processed internally.

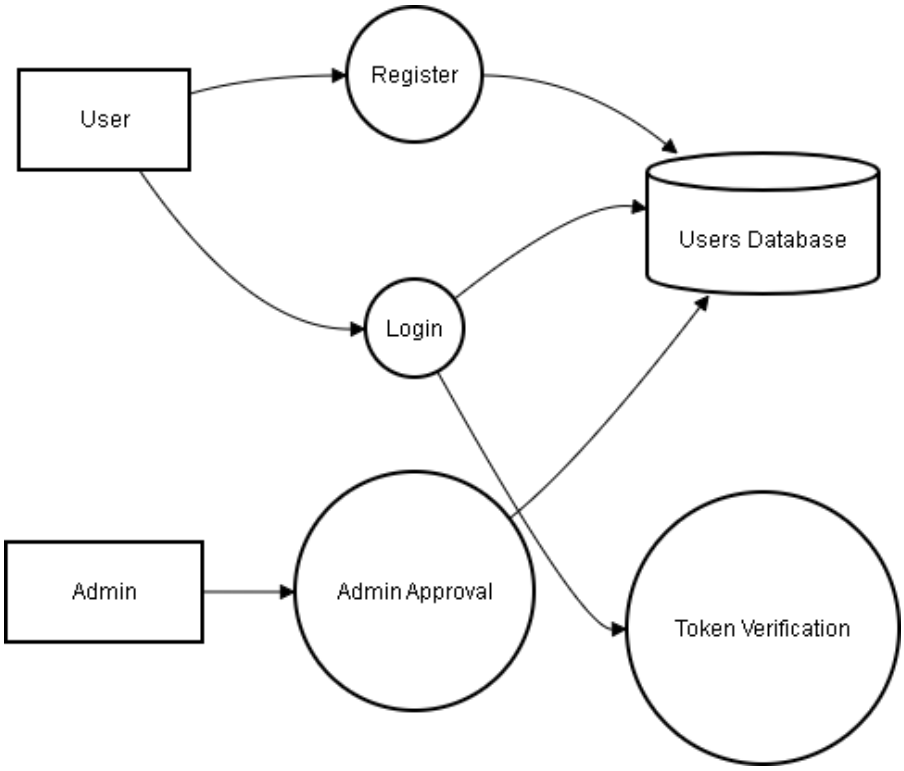
### 7.1 Authentication and Role Management

This module manages user authentication and role-based access control within the system.

The processes involved include:

• User Registration
• User Login
• Volunteer Application
• Admin Approval of Volunteers
• Token Verification for secure access

User credentials are stored securely in the Users database and authentication tokens are generated for secure communication between the client and server.



### 7.2 Disaster Monitoring and Alert System

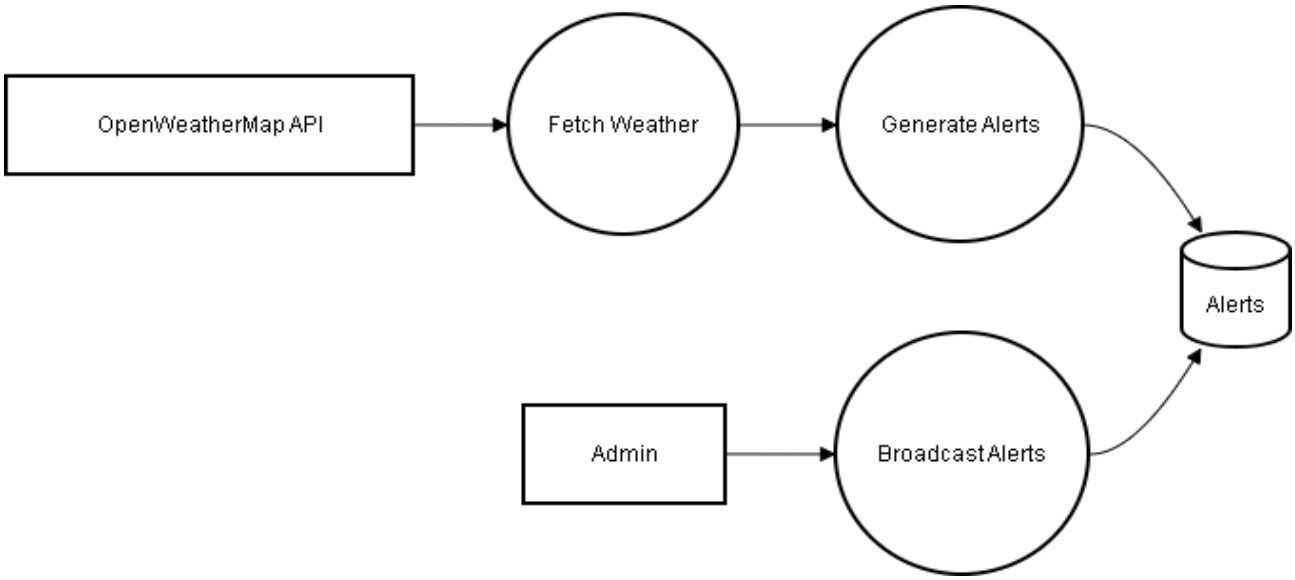
This module collects real-time weather data from the OpenWeatherMap API and generates disaster alerts based on rainfall conditions.

Administrators can also broadcast manual alerts to notify users about emergency situations.

The processes include:

• Fetch Weather Data
• Generate Rainfall Alerts
• Broadcast Emergency Alerts
• Notify Users

The generated alerts are stored in the Alerts database.



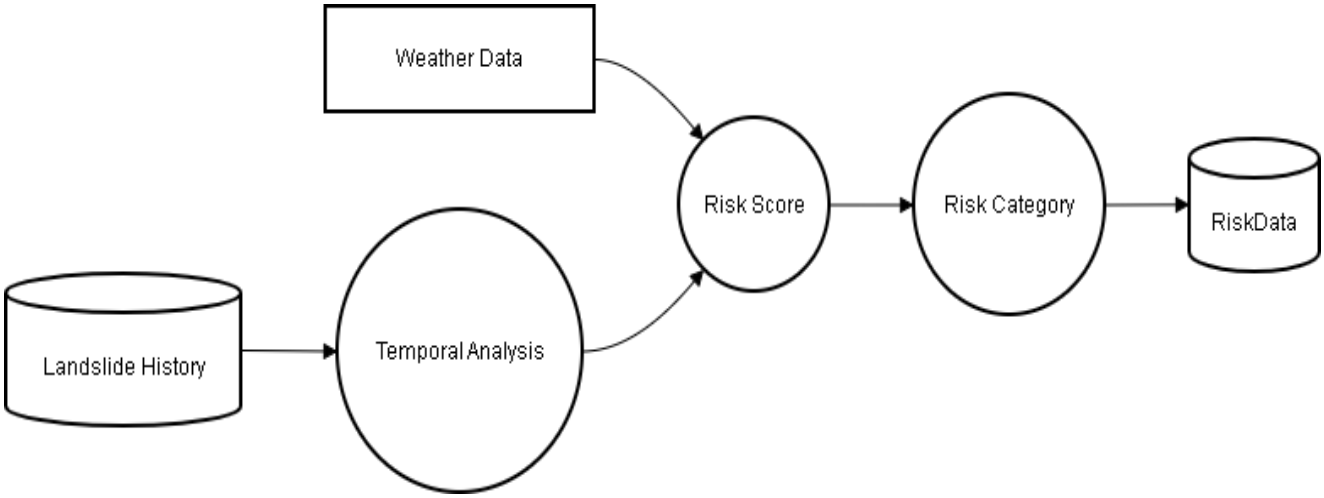
### 7.3 Risk Assessment and Analytics

The Risk Assessment module analyzes historical landslide data and current weather conditions to determine disaster risk levels for different districts.

The processes include:

• Loading historical landslide datasets
• Performing temporal analysis
• Calculating risk scores
• Categorizing risk levels

The results are stored in the RiskData database and used to display risk zones in the system.



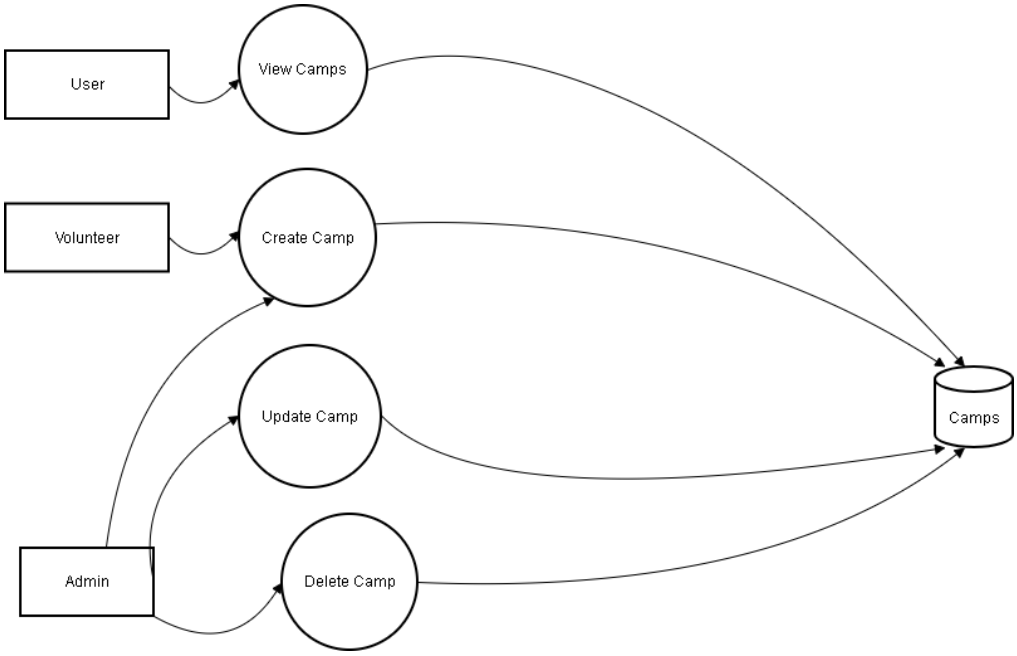
### 7.4 Camp Management System

This module manages relief camps used during disaster situations.

Administrators and volunteers can create, update, and delete camp records.

The processes include:

• Create Camp
• View Camp Details
• Update Camp Information
• Delete Camp




Camp information such as location, capacity,

and availability is stored in the Camps database.

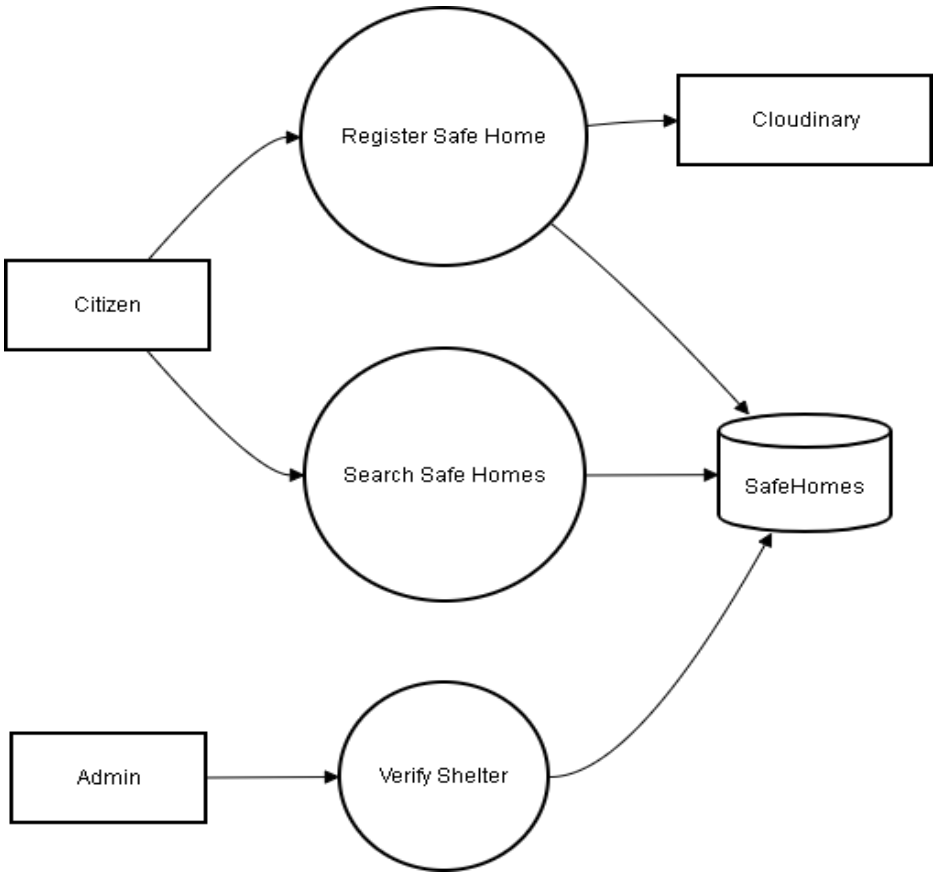
### 7.5 Safe Home and Shelter Management

This module allows citizens to register their homes as temporary shelters for disaster victims.

The processes include:

• Register Safe Home
• Admin Verification of Shelter
• Search Available Safe Homes
• Update Shelter Availability

Images of shelters are stored in Cloudinary, while shelter information is stored in the SafeHomes database.



### 7.6 Missing Person Tracking System

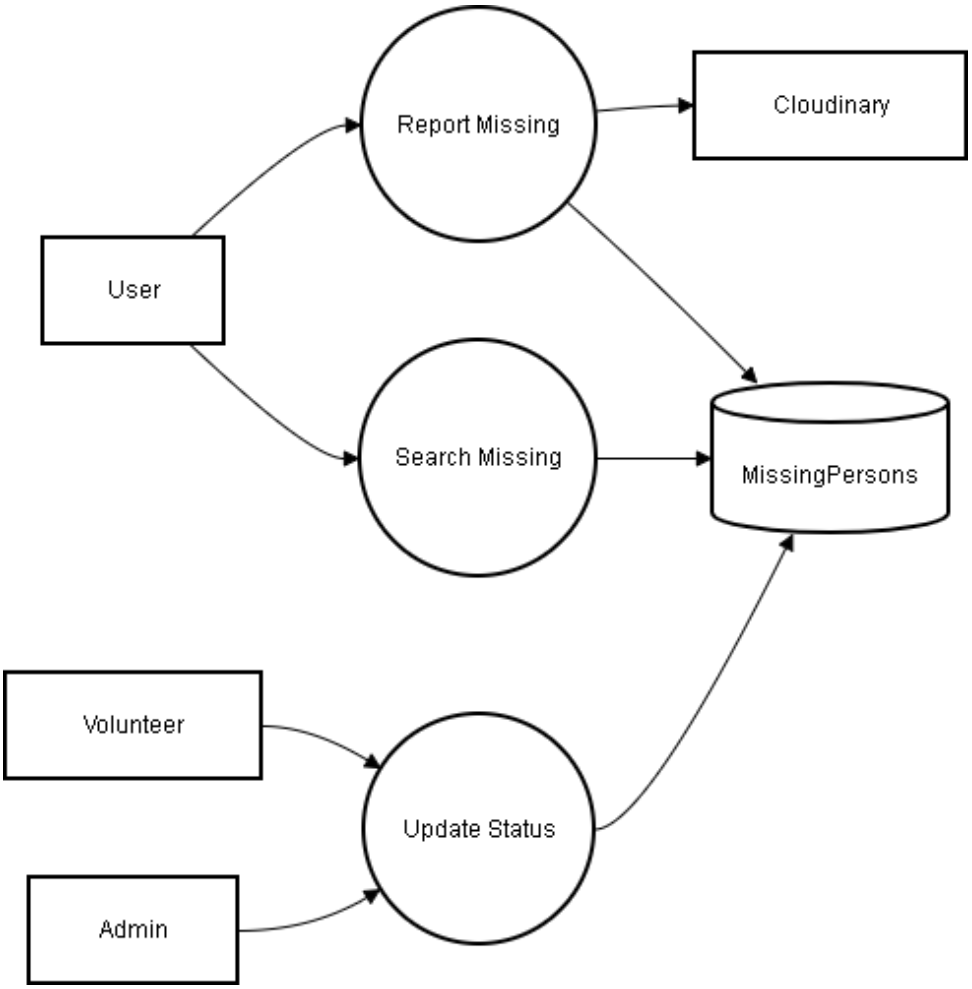
This module helps track missing individuals during disaster situations.

Users can report missing persons, upload photographs, and search for individuals.

The processes include:

• Report Missing Person
• Upload Missing Person Photo
• Search Missing Person
• Update Status of Missing Person

The information is stored in the MissingPersons database.



### 7.7 Volunteer Management System

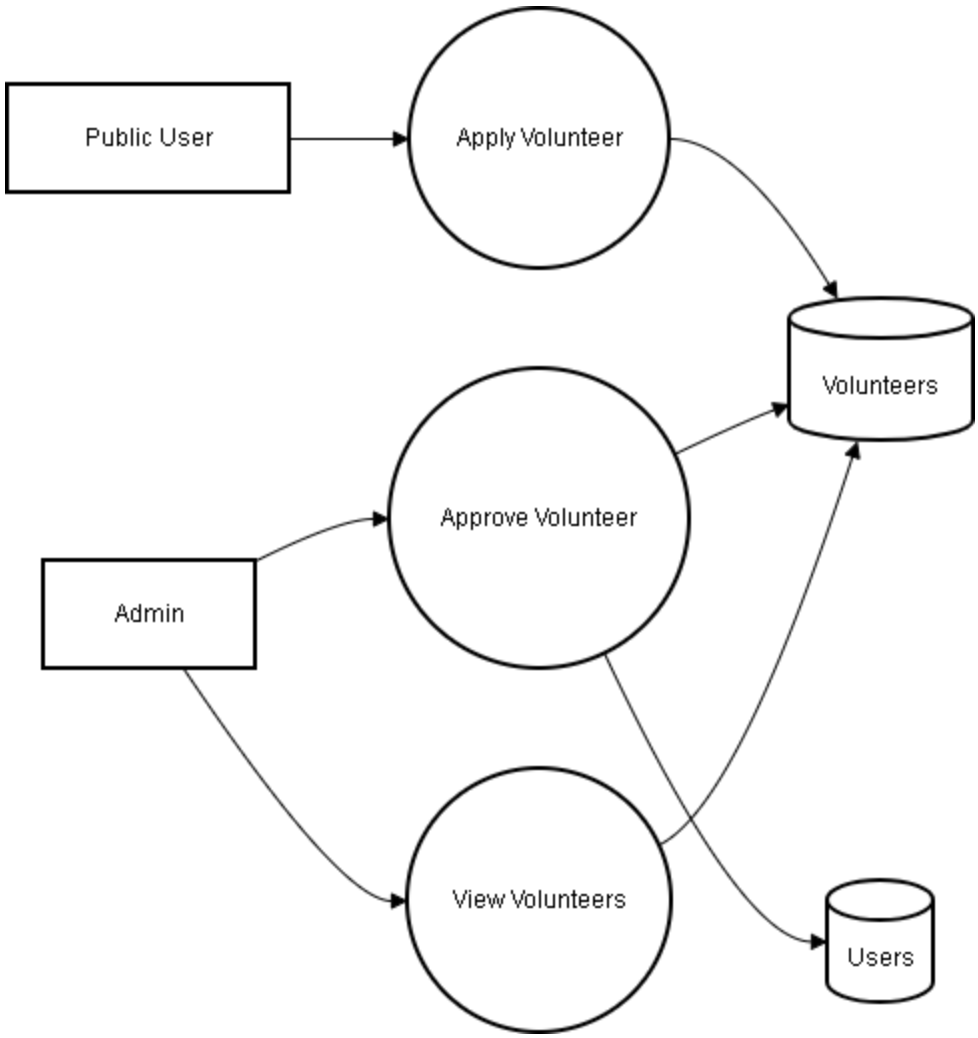
This module manages volunteer registrations and approvals.

Public users can apply to become volunteers, and administrators review and approve these applications.

The processes include:

• Register Volunteer
• Admin Approval
• View Volunteer List
• Update Volunteer Availability

Volunteer details are stored in the Volunteers database.



### 7.8 Donation Management System

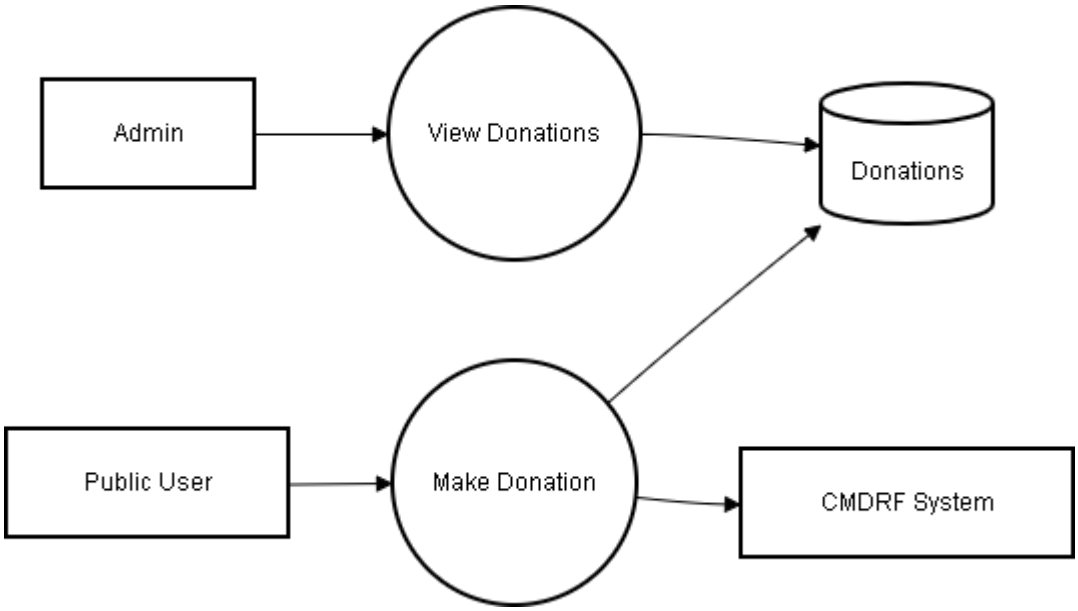
This module allows users to contribute donations to support disaster relief activities.

The donation system integrates with the Chief Minister's Distress Relief Fund (CMDRF).

The processes include:

• Display Donation Information
• Record Donation Details
• View Donation Records

Donation data is stored in the Donations database.




## 8 Summary

The Data Flow Diagrams presented in this chapter provide a clear representation of how data moves through the ResQ-DMS system. Starting from the high-level context diagram to detailed Level 2 diagrams, the diagrams help in understanding the interaction between users, processes, data stores, and external services.

These diagrams serve as an important part of the system design, ensuring that the system architecture is well structured and easy to understand.